North South University

Department of Electrical & Computer Engineering

Final Project Report

Project Name: HoldAString

GitHub Link: https://github.com/tjnblve-source/CSE299.17-Project-HoldAString-by-Ilma-Ibrahim-.git

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Introduction

In an increasingly digital world, the quality of human relationships is declining fast. While we have more ways to connect than we ever did in the past, the 'noise' of modern life leads to a social drift. ***HoldAString*** is a relationship management system designed to bridge this gap. By combining generative AI, Natural Language Processing, and mobile integrations, it acts as something more meaningful and efficient than a normal notes app.

Problem Statement

Social Problem

Although we have many ways to communicate in the modern times, the hassles of our own lives keep us away and distracted from our loved ones. Often times, even if we care about a person a lot, we fail to reach out to them. We may see a notification from them while we're busy and think, "I'll reply when I'm free," but the time never comes because nobody except our brain is there to remind us. And by the time we do remember, the context remains in the past and we cannot figure out how to start the conversation.

Technical Problem

While tools like notes app, To-Do list, calendars, etc. exist, they require a lot of manual labour from the user. We do not always have the energy to figure out exactly what to put down as the 'event' or 'details', and even if we just write down our memory, it can often end up seeming like a bunch of messy texts to our future self.

Moreover, regular technologies are passive. If a user sets a reminder, lives up to it, but then doesn't have any interaction with that person for a long period afterward, the 'relationship health' decays. In such a scenario, no existing app can figure out that a relationship is falling apart and remind the user accordingly. There is a lack of a system that can take a messy thought, turn it into an organized event, and then actively monitor if that relationship is being neglected over time.

Project Description

HoldAString is basically a digital safety net for a user's social life. Instead of just being another list of names and numbers, it's a system that actually pays attention to how well someone is keeping up with the people they care about.

It is built using a mix of web tools (PHP and MySQL), generative AI (Gemini API), NLP (Python spaCy), and a mobile app for the alerts.

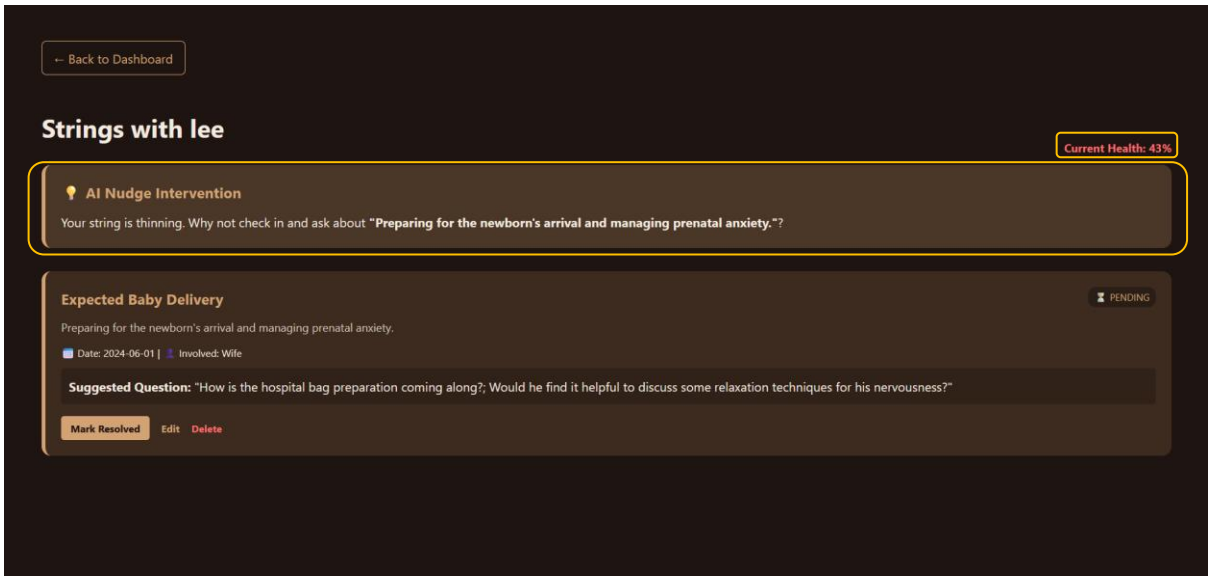
Here is a breakdown of how the different parts of the project work together to solve the problems mentioned earlier:

- **The String Health & Decay System:** The system stores connections in the form of 'strings'. Every string acts as a health bar. The initial health for each string starts at 100% and decays over time depending on the frequency of the user's interactions with the connection. Even the smallest interactions with the person, for example, a simple text or phone call, can earn health for the string. On the other hand, a whole week of interaction can take away 10% from the string health. It's a simple way of tracking who the user is accidentally ignoring before the connection fades away completely.
- **The 'Brain Dump', Processing, and Output:** One of the biggest issues with existing task apps is that the details have to be organized manually. To fix this problem, HoldAString uses an integrated system consisting of an API and an NLP library. The user can write a messy text description about what they talked about with their connection, for example, "I met Han today. He has a concert coming up, and he's quite stressed about it. His parents' anniversary is on 22nd May and he does not know what gift to get them." From this user input, the Gemini API will first extract each individual event and store them in a JSON array. Then, the spaCy NLP engine will extract the exact entity involved with the event and the date for the event. The exact attributes of the event will then be stored into the database, and the AI also produces contextual questions or conversation starters that the user can use while interacting with the connection next time.

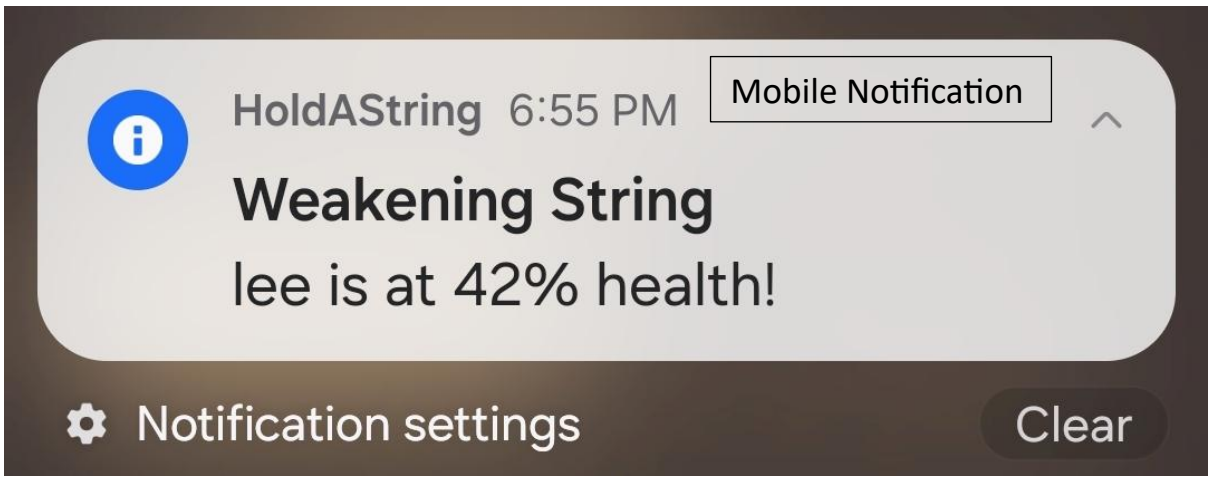
Extracted Events on String Dashboard

- **Smart Nudges & Android Integration:** As mentioned previously, each string is practically a health bar. Because the string health only decays every time

the user fails to interact with a connection for a week, the string health reaching below the critical point (50%) indicates that the user has already neglected their connection for at least 5 weeks. To remind the user that the relationship is drifting apart, the AI gives a nudge in the string dashboard and also sends a notification to phone when the user activates the app.

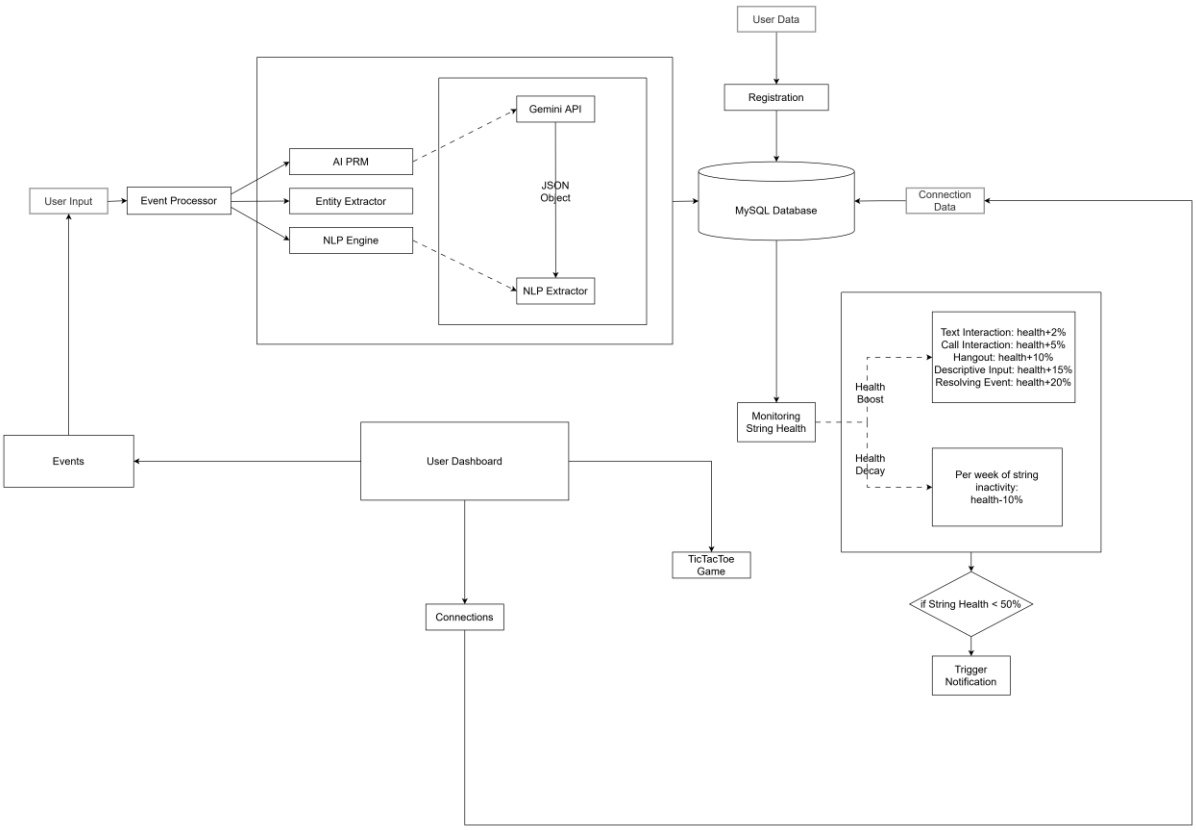


Alert on String Dashboard



Mobile Notification

Project Flowchart



Technologies Used

Database: MySQL

For the database, I used MySQL database. It uses four main tables.

- Users: To store user data achieved from each user registration.
- Connections: To store connection data given by user.
- Strings: To keep track of the health of each relationship and user's last interaction with the person.
- Events: To store events and details extracted by the API and NLP engine.

Backend: PHP

For the backend, using PHP allowed me to easily integrate the backend logic and processes with the frontend. When dashboard is loaded, PHP is the one talking to the database and fetching necessary data.

Frontend: HTML & CSS

To build the UI, I used HTML for the structure and CSS for the style. The UI is simple and user friendly.

Generative AI: Google Gemini API

I used the free Google Gemini Flash API because it's fast, at reading messy texts and can extract and parse the necessary details into JSON objects.

NLP Engine: Python & spaCy

SpaCy is a free, open-source Python library designed for advanced Natural Language Processing (NLP) in production environments. Because of its ability to do text preprocessing and Named Entity Recognition (NER), I used it as a "verification layer" for the event logging.

Mobile Integration: Android Studio & Java

To make the project more than just a website, I built an Android app wrapper.

- **WebView:** This acts as a window that displays the dashboard inside the app.
- **The Bridge (JavaScriptInterface):** This allows the website to talk to the phone's hardware. This is how the PHP code on the server is able to trigger native notifications on the user's phone.

<u>Cost Analysis (Real World Implementation)</u>

If I were to actually launch this for 1,000 users, the monthly cost would be surprisingly low because of the simplicity and efficiency of the tools.

- Server Hosting: A basic virtual server costs about $0.04 per hour. Over a month, that is ~$30.00 (To keep the PHP and Python code running 24/7).
- Database: Most cloud providers charge a flat monthly fee for a 'Managed Database'. It covers the storage of the tables (Users, Connections, Strings, Events) and makes sure the data doesn't get corrupted if the server crashes. The monthly cost would be around ~$25.00 (To safely store everyone's connections).
- AI Costs: Google charges for Gemini API by the 'token' (basically by the word). Gemini-Flash is their most affordable model. If 1,000 users each do 2 'Brain Dumps' a week, that makes 8,000 requests a month. At current prices, that's actually less than $10, so $15 gives us a safe cushion for longer notes. So the final monthly cost would be ~$15.00.

Total Cost: About $70.00 per month for 1,000 users. Or, ~$0.07 per user.

Conclusion

HoldAString successfully demonstrates how we can use the same technology that usually distracts us to actually bring us closer to people. By turning the abstract idea of 'drifting apart' into a math-based health bar and using AI and NLP to handle the 'manual labour' of organizing notes, the project provides a practical safety net for our social lives.

While the current version works great as a prototype, future versions could include direct integration with messaging apps (like WhatsApp) to log interactions automatically. Overall, the project proves that with the right tools, we can make sure our most important connections don't fade away just because life gets busy.